

Mohammad Rafay

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EDUCATION

Texas A&M University <i>Bachelor of Science in Computer Engineering</i>	College Station, TX December 2025
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CERTIFICATIONS

CompTIA Information Technology+ (COMP001021854961)	May 2021
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EXPERIENCE

Software Engineer <i>Norstella</i> - Designed and implemented Java Spring Boot backend services and REST APIs for a healthcare platform, supporting secure client-server workflows, service-to-service communication, and reliable data processing across distributed application components. - Optimized SQL queries and backend data access patterns, reducing slow query execution time and improving reliability across REST API workflows. - Contributed to Agile development workflows by managing code in Azure DevOps Git repositories and supporting unit testing and code reviews prior to submission, strengthening release stability and code quality. - Developed, deployed, and tested Java Spring Boot services in an AWS-hosted virtual machine environment, using remote server workflows, API validation, logging, and debugging to support cloud-based backend development. - Applied AI-assisted development tools (Cursor/Forge AI) to streamline backend engineering and testing workflows, supporting faster debugging, code refinement, and more efficient validation of Java Spring Boot services.	January 2026 – Present <i>Richmond, Texas</i>
Software QA Engineer - Career Experience (Messages App) <i>Apple Inc.</i> - Designed and implemented automated test scripts in Python and Swift using XCTest for iOS/macOS Messages App, developing test cases and coverage strategies at scale to validate UI workflows and interactions across software releases. - Maintained test development roadmaps in collaboration with cross-functional engineering teams, translating software requirements into testable criteria and driving quality ownership across customer-facing applications. - Built and optimized reusable test frameworks to streamline automation pipelines, integrate results into CI/CD reporting cycles, and support scalable regression testing across software releases. - Executed functional, regression, and performance testing, triaged and debugged critical issues, and delivered actionable reports that improved system performance and helped maintain stable software releases.	January 2025 – June 2025 <i>Cupertino, California</i>
Software Engineer - Internship <i>Y-12 National Security Complex</i> - Built an interactive route planning simulation using Cesium, Unity, and C#, allowing users to dynamically create the most efficient routes for transporting nuclear materials while ensuring maximum security. - Implemented cybersecurity policies to strengthen the software for the company's alarm response training, enhancing the facility's protection against imminent threats by 40%. - Accelerated training throughout the organization by 20% developing modules to counteract cyber threats in resource transportation.	May 2024 – January 2025 <i>Oak Ridge, Tennessee</i>
Cybersecurity Engineer - Internship <i>Texas Department of Transportation</i> - Maintained Security Policy content for compliance and accuracy throughout the organization. - Conducted Third Party Risk Management assessments for enhanced protection of management systems. - Utilized KQL within Microsoft Azure, to analyze research metrics, supporting data-driven decision-making and strategy formulation.	May 2023 – August 2023 <i>Austin, Texas</i>

PROJECTS

Accent Auto (Client Website) <i>HTML, CSS, JavaScript, Node.js, Express</i> accentautohtx.com Built a client website using HTML, CSS, JavaScript, Node.js, and Express, implementing responsive inventory pages, vehicle detail views, status filters, and customer contact and inquiry workflows including form submission and email integration.	Jan. 2026 – Present
BluStick <i>TypeScript, JavaScript, Swift, C++, SQL</i> Developed a hardware-software system that collects Bluetooth signals from distributed sensors, performs real-time triangulation, and visualizes device locations on an interactive and responsive cross-platform user interface.	August 2025 - December 2025
LED Matrix Camera Display <i>Python</i> Built an interactive LED matrix system that streams live webcam input and button-triggered patterns, validating UI responsiveness and ensuring seamless interaction between input devices and visual output.	January 2024 - May 2024
Point-of-Sale System <i>TypeScript, JavaScript, Java, HTML, React</i> Developed a React and TypeScript point-of-sale application with role-based employee, manager, and cashier interfaces, enabling order management, page-to-page workflows, and structured front-end interactions.	January 2024 – May 2024

SKILLS

Languages: Python, Java, SQL, C++, C#, Swift, Bash, TypeScript, JavaScript, HTML, CSS, Haskell
Technologies: Spring Boot, REST APIs, JSON, Azure DevOps, AWS Virtual Machine Environment, SaaS Platforms, Postman, Linux/Unix, macOS, Windows, IntelliJ, VS Code, JIRA, Agile, Cursor AI
Soft Skills: Communication, Cross-Functional Teamwork, Problem Solving, Analytical Thinking, Time Management, Leadership

LEADERSHIP POSITIONS

President Corporate Chair <i>Honors Society for Electrical & Computer Engineers (HKN)</i>	January 2023 – January 2025
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